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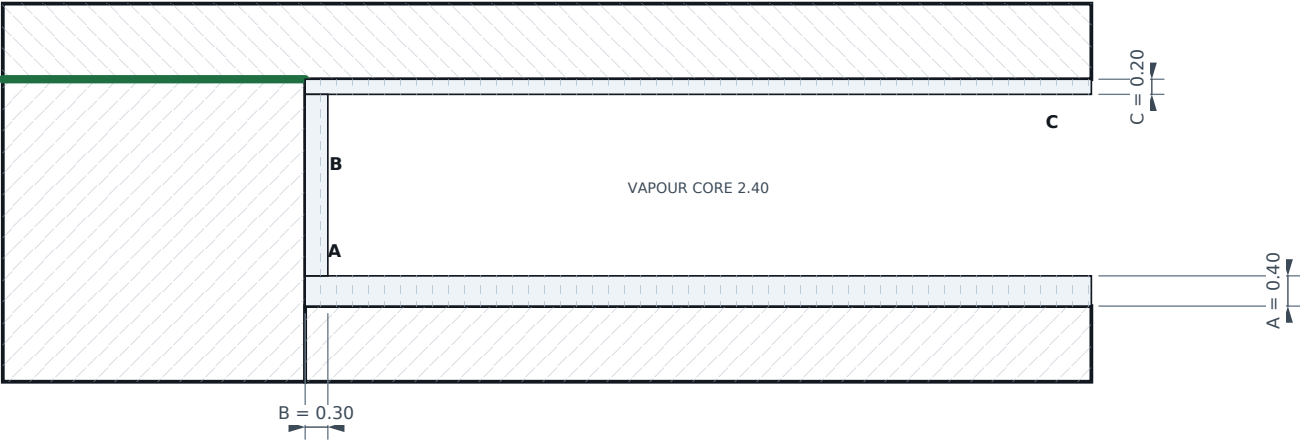
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WICK ZONE SECTION

SCALE 10:1 — ASSEMBLED



Zones A, B and C are sintered in ONE cycle and must be capillary-continuous:

condensate returns C → B → A. A break at the B/A junction causes evaporator dry-out.

| WICK ZONES |                             |      |                            |
|------------|-----------------------------|------|----------------------------|
| ZONE       | LOCATION                    | THK  | FUNCTION                   |
| A          | Pocket floor, around posts  | 0.40 | evaporator + liquid supply |
| B          | Pocket side walls (4 off)   | 0.30 | perimeter return path      |
| C          | Lid underside, posts masked | 0.20 | condensate collection      |

| SINTER PROCESS                     |                         |                                   |
|------------------------------------|-------------------------|-----------------------------------|
| PARAMETER                          | VALUE                   | TOL / NOTE                        |
| Cu powder, spherical, gas-atomised | 45 - 75 um              | sieved, -200/+325 mesh            |
| Effective pore radius              | 12.6 um                 | = 0.21 x d(particle)              |
| Porosity                           | 55 %                    | ± 5 %, quantitative metallography |
| Permeability (Blake-Kozeny)        | 1.97e-11 m2             | derived, not measured             |
| Capillary head, water @ 60 C       | 10.5 kPa                | vs 0.66 kPa gravity head          |
| Sintering temperature              | 950 C                   | ± 10 C                            |
| Dwell                              | 30 min                  | at temperature                    |
| Atmosphere                         | H2 (dew pt < -40 C)     | or vacuum < 1e-3 Pa               |
| Ramp / cool                        | 10 C/min / furnace cool | no quench                         |
| Tooling                            | graphite mandrel + mask | masks post tops, rim, fill port   |

WICK ACCEPTANCE

| ACCEPTANCE TEST       | REQUIREMENT       | METHOD                         |
|-----------------------|-------------------|--------------------------------|
| Porosity              | 55 % ± 5          | image analysis, 3 sections     |
| Wick thickness A      | 0.40 ± 0.05       | cross-section, 5 points        |
| Capillary rise, water | > 25 mm in 60 s   | witness coupon, vertical       |
| Groove continuity     | open, no blockage | back-light / CT, first article |

POWDER TRADE STUDY — WHY 45-75 um

| POWDER                    | r eff   | dP cap   | Q cap VERT | VERDICT           |
|---------------------------|---------|----------|------------|-------------------|
| 25 um                     | 5.2 um  | 25.2 kPa | 348 W      | clogs, low K      |
| 45 - 75 um                | 12.6 um | 10.5 kPa | 804 W      | SELECTED          |
| 100 um                    | 21.0 um | 6.3 kPa  | 1279 W     | coarse, ok        |
| 150 um                    | 31.5 um | 4.2 kPa  | 1802 W     | marginal tilt     |
| 400 um (original drawing) | 100 um  | 0.4 kPa  | 0 W        | FAILS — < gravity |

The original 200 um pore spec gives 380 Pa of capillary head against a 500 Pa gravity head

over 68 mm: the wick cannot lift liquid at all in any vertical orientation.

CAPILLARY PRESSURE BUDGET — WORST CASE, VERTICAL

| TERM                               | VALUE       | NOTE                      |
|------------------------------------|-------------|---------------------------|
| Capillary head available 2σ/r      | 10 508 Pa   | water @ 60 C, r = 12.6 um |
| Liquid loss through wick + grooves | – 12.6 Pa/W | at 804 W: 10 127 Pa       |
| Vapour loss through 2.40 core      | – 0.2 Pa/W  | negligible                |
| Gravity head, 68 mm adverse        | – 656 Pa    | vertical, evaporator up   |
| Margin at 113 W design point       | + 8 428 Pa  | 7.4 x                     |

The wick is sized by the vertical case. Horizontal operation carries 858 W and is never the limit.

PROCESS NOTES

- Mask post top faces and the 4.0 rim before sintering — sintered powder on a bond land destroys hermeticity.
- Mask the fill port bore. Powder in the pinch-off tube prevents a gas-tight cold weld.
- Grooves must remain open under the wick. Verify by back-light or CT on the first article.
- Sinter and diffusion bond may share one furnace cycle only if the bond fixture applies load after the sinter dwell; otherwise run separately.
- No organic binders. Any residue outgasses into the sealed chamber as non-condensable gas.

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE

SINTERED WICK — PROCESS SPECIFICATION

IN-SITU SINTERED Cu POWDER / ALL WICK ZONES

SCALE

10:1

UNITS

mm

THIRD ANGLE PROJECTION

DRAWING No.

VC-103

REV

A

SHEET

4 / 7

DRAWN

AI-ASSISTED DRAFT

CHECKED

— PENDING —

MATERIAL

SEE BOM

FINISH

SEE NOTES

DATE

2026-09-09